TigerAware Assistant: A New Serverless Implementation of Conversational Agents for Customizable Surveys on Smart Devices

Yohanes Handrianto, Rui Huang, and Yi Shang
Department of Electrical Engineering and Computer Science (EECS)
University of Missouri
Columbia, Missouri, USA
yph374@mail.missouri.edu, rhhq7@mail.missouri.edu, shangy@missouri.edu

***Abstract*—Applications of conversational agents, also called chatbots, have accelerated in recent years. However, it is difficult for people with limited programming skills to implement chatbots and analyze the data collected by chatbots. In this paper, we present a new serverless implementation of conversational agents for delivering customizable surveys on smart devices, including smartphones and smart speakers, and a web-based system for applying advanced analytics techniques to survey data using cloud services. The system is called TigerAware Assistant, as part of the TigerAware platform for customizable mobile survey and sensor data collection. TigerAware Assistant enables non-technical people to easily create and deploy chatbots on various mobile devices to conduct surveys through conversations in natural languages via auditory and textual methods. It also provides a web-based system for survey data visualization, statistical analysis, and advanced machine learning-based data analysis using cloud services on survey responses collected by chatbots.**

Keywords-Chatbot, Conversational Agent, Mobile Survey, Cognitive AI, Cloud Function, Serverless Computing

I. INTRODUCTION

For a long time, researchers in many disciplines have been using surveys to collect data for their research. From psychology to marketing, researchers are looking for ways to easily deploy survey-based studies, which can help them make insightful decisions. Well-known web-based survey systems include Survey Monkey (https://www.surveymonkey.com/) and Qualtrics Survey (https://www.qualtrics.com/). For conducting mobile surveys, an example commercial system is REDCap (https://projectredcap.org/).

Due to the proliferation of smart mobile devices, such as smartphones and smart speakers, and the rapid advances in AI, conversational agents, also called chatbots, are becoming an attractive tool for conducting surveys [1-5]. A chatbot is a software program that interprets and responds to statements made by users in natural languages. However, there is no easy way for people with limited programming skills to implement chatbots and analyze the data collected by chatbots. To develop survey chatbots for their particular studies, researchers first design the survey questions for the chatbots and decide which platforms to deploy the chatbots. Then, they have to hire programmers to implement the chatbots, which can take months and be quite expensive. Once the chatbots are implemented and deployed, the survey questions (i.e. conservations) cannot be modified. To analyze the survey data collected by chatbots, researchers have to download the data and use some existing data analysis software offline. In this approach, it's impossible for researchers to perform real-time data analysis and modify their survey questions adaptively, during a field study, to better achieve their research goals.

In this paper, we present a new serverless implementation of conversational agents for delivering customizable surveys and a web-based system for analyzing survey data using cloud services. The system is called TigerAware Assistant, as part of the TigerAware platform for customizable mobile survey and sensor data collection [6]. TigerAware Assistant enables researchers and people without any programming knowledge to easily create and deploy chatbots on various mobile devices, such as iPhones, Android smartphones, and smart speakers, to conduct surveys through conversations in natural languages. Survey questions and survey structures could be modified by researchers on the fly during field studies. TigerAware Assistant also provides a web interface for data visualization, statistical analysis, and advanced machine learning-based data analysis through cloud services on survey responses collected by chatbots. Using this system, researchers could visualize and analyze survey data collected by chatbots in real-time and modify their surveys adaptively, during a field study, for better results.

II. RELATED WORK

Woebot is a Web-based cognitive-behavioral therapeutic (CBT) application [5]. It implemented a fully automated conversational agent to deliver a self-help program for college students who self-identify as having symptoms of anxiety and depression. It offered a convenient, engaging way of getting support at any time and demonstrated that chatbots appear to be a feasible, engaging, and effective way to deliver CBT. In [7], we proposed a chatbot for healthy coping with diabetes. The Google Home app was developed to assist elderly patients with self-management of type 2 diabetes. This app combined the voice interface of Google Home for hosting the conversational agent and a web interface for data visualization to reduce the burden of monitoring diabetic consequences for the user.

TigerAware [6] is an easy-to-use customizable mobile survey and sensor data collection and analysis system. It is designed for researchers in different disciplines to easily create and deploy mobile surveys and has been successfully used in several psychology studies with sophisticated mobile surveys, including surveys with branching structures, daily

978-1-7281-4127-5/19/$31.00 ©2019 IEEE
DOI 10.1109/TransAI46475.2019.00023

Authorized licensed use limited to: Sungkyunkwan University. Downloaded on April 11,2026 at 09:35:55 UTC from IEEE Xplore. Restrictions apply.

scheduled surveys, random surveys, follow-up surveys, and sensor data collected from both smartphone built-in sensors and external wireless sensors. TigerAware architecture consists of three main components: 1) a web dashboard for user administration, survey creation, survey data visualization, and analysis, 2) a Firebase server for storing user information, survey blueprints and data, and sensor data, and 3) Android and iOS mobile apps for conducting surveys and data collection. In [8], a data analysis module to TigareAware was proposed, which includes various survey response analysis, ranging from simple statistical analysis on numerical data to cutting-edge deep learning models on image, audio, and text data. As a proof of concept, some existing pre-trained deep learning models on image and audio were used to predict the emotional state of a user. The proposed TigerAware Assistant is a new addition to TigerAware to provide chatbots capability on popular mobile platforms including iOS devices, Android devices, and smart speakers, and improved data visualization and analysis methods.

III. TIGERAWARE ASSISTANT DESIGN AND IMPLEMENTATION

TigerAware is being developed to become a comprehensive mobile survey platform that supports both mobile apps and conversational agents, and provides some web-based survey data visualization and analysis tools. The proposed TigerAware Assistant and web-based data analysis engine are new additions to the existing TigerAware platform.

Fig. 1 shows the TigerAware system architecture. Its central database is a NoSQL database containing three major parts: a survey blueprint store, a survey response data store, and a user account information store.

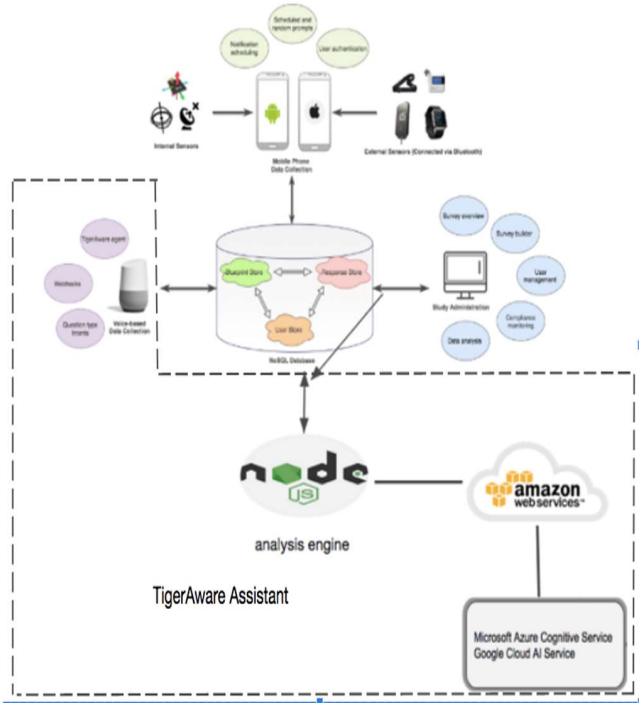

Figure 1. TigerAware System Architecture.

TigerAware Assistant is the part within the dash lines in Fig. 1. It can create chatbots that deliver surveys through conversations. It uses the blueprint store to hold instructions for conversations, such as the conversation name and instructions for what to do in each step of a conversation. It uses the data store to keep track of conversation responses of the survey takers from mobile devices. The storage of response data alongside the corresponding blueprint is important because the blueprint enables appropriate interpretation of the conversation data.

TigerAware Assistant is developed based on serverless and Function-as-a-Service (FaaS) cloud computing paradigm [9-11], allowing software developer to develop, run, and manage system functionalities without the complexity of building and maintaining their own infrastructure that is typically associated with developing and launching mobile apps and their associated servers. In serverless computing, cloud providers run the servers and dynamically manages the allocation of machine resources. Scaling, capacity planning, and maintenance operations are handled by cloud providers.

TigerAware Assistant is implemented as a cloud function application hosted on Firebase Cloud Function Service. Firebase Cloud Function and node.js based backend provide all the necessary REST APIs, which allow for data to be retrieved, saved, updated, and transformed in the conversational agent. TigerAware Assistant uses Google's Dialogflow for natural language conversations. Firebase cloud functions are implemented as the fulfillment of the Dialogflow platform. Fulfillment is code that is deployed as a webhook to let Dialogflow agent call logic on an intent-by-intent basis. During a conversation, fulfillment allows users to use the information extracted by Dialogflow's natural language processing function to generate dynamic responses or trigger actions on the back-end. Building chatbots this way allows TigerAware Assistant to generate dynamic responses based on survey answers and endpoint services, and enables researchers to implement customized rules for their studies. TigerAware Assistant cloud functions can utilize serverless NoSQL databases, such as Firebase, that lets researchers store and sync data from their survey participants in real-time.

The design of TigerAware Assistant is highly modular and generic, giving researchers the flexibility to create, edit and deploy conversational agents, and developers the flexibility to modify and add functionality to their programs easily. It delivers surveys as conversations to participants, and collecting and saving responses to a cloud real-time database, such as Firebase.

Fig. 2 illustrates the TigerAware Assistant workflow. It consists of five major components:

1) A smart device or smart speaker, such as Google Home. The device enables users to speak in natural languages to interact with TigerAware surveys through Google Assistant.

2) Google Assistant, an AI-powered virtual assistant developed by Google that is available on many smartphones and smart home devices. Google Assistant runs on Google Home and enables it to communicate with a user in natural language.

3) Dialogflow, an end-to-end, build-once deploy-everywhere development suite for creating conversational

Authorized licensed use limited to: Sungkyunkwan University. Downloaded on April 11,2026 at 09:35:55 UTC from IEEE Xplore. Restrictions apply.

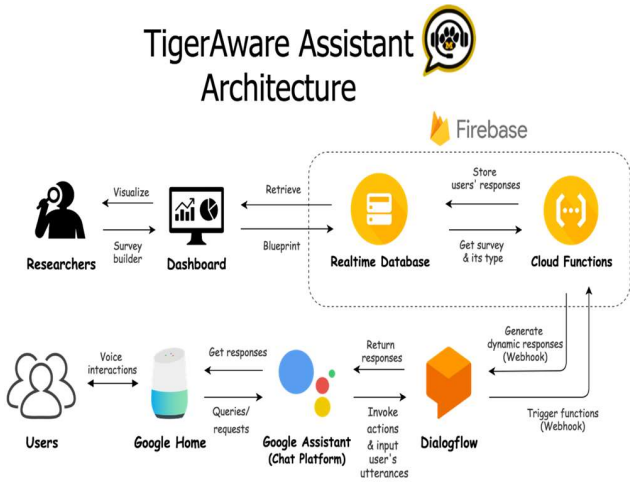

Figure 2. TigerAware Assistant workflow.

agents from Google (https://cloud.google.com/dialogflow/). It is powered by Google machine learning to recognize a user's intent, which categorizes an end-use's intention for one conversation turn. In order to create an agent, Dialogflow needs to access survey data through cloud functions. Each agent has multiple intents to handle different aspects of a conversation. When a survey participant says something, Dialogflow matches the expression to the best intent and executes its corresponding action. The actions are implemented using Google Cloud Functions.

4) Google Cloud Functions, Google's cloud computing service. Google Cloud Functions support serverless computing by allowing users to develop, run, and manage application functionalities without the complexity of building and maintaining their own computing infrastructure. Cloud functions are used to deliver surveys retrieved from Firebase to Dialogflow, and to provide customized actions of intents.

5) A web dashboard for researchers to build and deploy surveys, manage participants, and visualize and analyze survey responses.

As shown in Fig. 2, when researchers create surveys on the web dashboard, survey blueprints containing survey questions and structures are pushed to the Firebase database. Then, survey participants could activate a specific survey by the survey name on a Google Home device. Users can talk to Google Home with Google Assistant running to conduct the survey as a conversation. Google Assistant then sends requests to Dialogflow, which triggers cloud functions to fulfill the requests and generate conservation responses based on the survey blueprint.

Fig. 3 shows an example of survey conversions with a TigerAware Assistant chatbot. Saying the survey name "pizza" started the conversation. The first question, "Do you like pizza?" is a yes/no question, whereas the second question is a multiple-choice question. The choices are listed by the chatbot for the user to choose from.

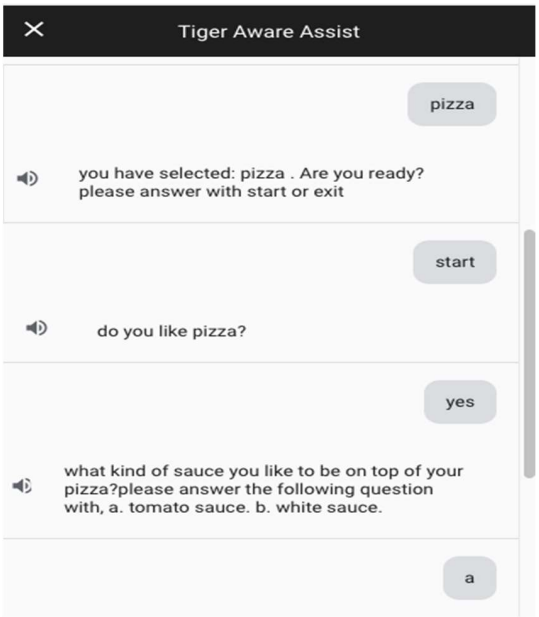

Figure 3. A conversation example with a TigerAware Assistant chatbot about pizza.

TigerAware Assistant can also answer location and weather questions by accessing third-party web services. For example, to answer a weather question, Google Cloud Function sends a request with the location and time/date to a third-party weather service. This weather service returns a webhook response in JSON format. Cloud function custom fulfillment parses the JSON data and delivers a response to the user with the relevant information. An example of a conversation with these questions is shown in Fig. 4.

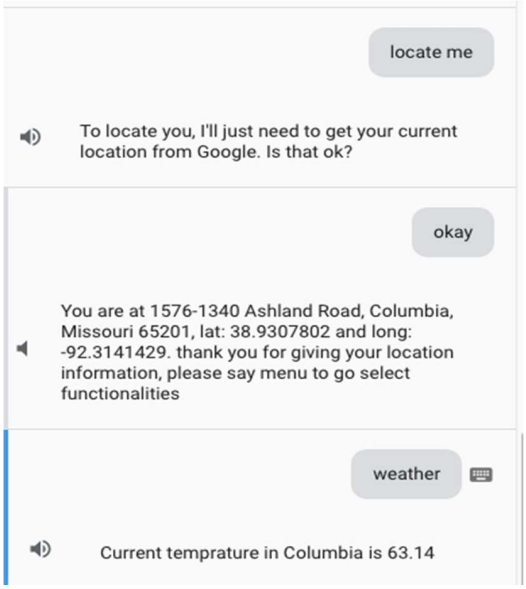

Figure 4. A conversation example with a TigerAware Assistant chatbot about location and weather.

Authorized licensed use limited to: Sungkyunkwan University. Downloaded on April 11,2026 at 09:35:55 UTC from IEEE Xplore. Restrictions apply.

IV. Web-based Data Analysis Engine Design and Implementation

The web-based data analysis engine is designed to add easy-to-use analysis tools to TigerAware for researchers to perform some quick data visualization and analysis.

The analysis engine pulls survey data from Firebase and calls implemented analysis functions, ranging from common statistical functions to advanced deep-learning-based image and text recognition tools provided by Google Cloud AI Service (https://cloud.google.com/products/ai) and Microsoft Azure Cognitive Service (https://azure.microsoft.com). A data analysis web interface is developed for researchers to choose various analysis functions and visualize results.

Fig. 5 shows the system architecture of the TigerAware analysis engine. It has four main components:

- Analysis web page for researchers to interactively apply available analysis methods on their survey data.
- Data storage, a Firebase database containing data including survey questions, survey responses, analysis results, etc.
- Analysis engine, a RESTful service that receives requests from the analysis web page, calls corresponding analysis programs, and stores analysis results in Firebase. The analysis functionalities include basic statistical functions and advanced deep learning analysis, such as emotion and object detection on images, sentiment analysis on text, content classification on text, etc.
- Third-party analysis services, including Microsoft Azure Cognitive Service and Google Cloud AI Service, for analyzing complex data, such as image and text.

Fig. 6 shows an example of the emotion detection result returned by Azure Cognitive Service on a face image. The service classifies emotions into eight categories: anger, contempt, disgust, fear, happiness, neutral, sadness and surprise. In this example, the detection result is happiness.

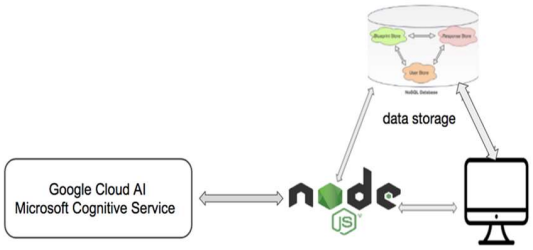

Figure 5. System architecture of the Web-based analysis engine.

Figure 6. An example of the emotion detection result returned by Microsoft Azure Cognitive Service on a face image.

In conclusion, the new TigerAware Assistant and web-based data analysis engine accomplished the following:

1) Significantly reduce the software development time and cost for delivering surveys on a wide-range of smart devices as chatbots. For example, we were able to create a cognitive behavior therapy chatbot similar to Woebot in a few hours.
2) Enable non-programmers to create generic customizable chatbots with rich features that could be used in many applications.
3) Support real-time survey data visualization and analysis.
4) Enable researchers to control and modify their chatbots easily without changing software code or redeployment of chatbots.
5) Enable researchers to access and use advanced data processing tools offered in AI cloud services, such as Google Cloud AI Service and Microsoft Azure Cognitive Service, to process and analyze the data collected by their chatbots.

References

[1] B.A. Shawar and E. Atwell, "*A comparison between Alice and Elizabeth chatbot systems,*" School of Computing Research Report, University of Leeds, 2002.

[2] M. Dahiya, Menal, "A Tool of Conversation: Chatbot," *International Journal of Computer Sciences and Engineering*, 5(5):158-161, May 2017.

[3] B. Setiaji and F. W. Wibowo, "Chatbot Using a Knowledge in Database: Human-to-Machine Conversation Modeling," *International Conference on Intelligent Systems, Modelling and Simulation*, 2016.

[4] S.A. Abdul-Kader and J.C.Woods, "Survey on Chatbot Design Techniques in Speech Conversation Systems," *International Journal of Advanced Computer Science and Applications,* 6(7), July 2015.

[5] K.K. Fitzpatrick, A. Darcy, and M. Vierhile. "Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial," *JMIR Mental Health,* April-June, 4(2):e19, 2017.

[6] W. Morrison, L. Guerdan, J. Kanugo, T. Trull and Y. Shang, "TigerAware: An Innovative Mobile Survey and Sensor Data Collection and Analytics System," *IEEE International Conference on Data Science in Cyberspace*, 2018.

[7] A. Cheng, V. Raghavaraju, J. Kanugo, Y. P. Handrianto, and Y. Shang, "Development and evaluation of a healthy coping voice interface application using the Google home for elderly patients with type 2 diabetes," *IEEE Consumer Communications and Networking Conference*, 2018.

[8] J. Rogers, D. Simmons, M. Shah, C. Rowland, and Y. Shang, "Deep Learning at Your Fingertips", *IEEE Consumer Communications and Networking Conference*, 2019.

[9] L.F. Albuquerque Jr., F.S. Ferraz, R. Oliveira, and S. Galdino, "Function-as-a-Service X Platform-as-a-Service: Towards a Comparative Study on FaaS and PaaS," *Twelfth International Conference on Software Engineering Advances*, 2017.

[10] M. Yan, P. Castro, P. Cheng, and V. Ishakian. "Building a Chatbot with Serverless Computing," *1st International Workshop on Mashups of Things and APIs*, 2016.

[11] P. Leitner, E. Wittern, J. Spillner, and W. Hummer, "A mixed-method empirical study of Function-as-a-Service software development in industrial practice," *Journal of Systems and Software*, Volume 149, Pages 340-359, 2019.

Authorized licensed use limited to: Sungkyunkwan University. Downloaded on April 11,2026 at 09:35:55 UTC from IEEE Xplore. Restrictions apply.